SUPERVISOR DISCUSSION NOTE / 8 OCTOBER 2026

Lumori

FYP Scope, Target Users and Expected Value

Lumori is an AI-assisted system for creating and reviewing educational content from course materials. It combines retrieval-augmented generation (RAG), structured authoring and traceable sources to support explanations, quizzes and assignments. The project aims to make material preparation and self-practice more convenient while keeping content open to inspection and revision.

1. Target users and initial setting

User group	Main tasks
Primary: university educators and teaching assistants	Prepare explanations, formative quizzes and assignments; inspect sources and answers; revise and export materials.
Secondary: undergraduate students	Create revision materials from their own course notes, practise with answers and explanations, or use materials distributed by educators.

The proposed initial evaluation will focus on one undergraduate computing course, such as Data Structures. This bounds the study; it does not establish suitability across all subjects.

2. Core project scope

Course materials and retrieval. Import and parse course sources, retrieve relevant passages and retain source references for later checking.

Content and assessment generation. Produce explanations, quizzes and assignments with answers and explanations, guided by learning objectives, prior knowledge, language, question format and intended difficulty.

Review and revision. Support source inspection, AI-assisted checks, manual confirmation, item-level edits and version history. Distinguish AI review from human confirmation and reassess revised content.

Two additional output modalities. Create audio explanations and visual teaching aids from reviewed text; check each modality for consistency with its source version.

Learning access and delivery. Provide reading and self-practice views, answer reveal and document export. Educators can distribute exported materials through existing teaching channels.

System evaluation. Assess correctness, source support, intended difficulty, multimodal consistency, usability, completion time and available cost information.

3. Optional extension: AI voice discussion

If time permits after the core functions and required evaluation, users will be able to ask spoken questions and follow up on selected course materials or generated content. Responses will combine speech with text and relevant source references. Discussion-based edits will remain suggestions until accepted. Unlike audio playback, this is an interactive feature. It is not required for core project completion.

What changes for users?

The main changes concern preparation, revision and learning support. Educators continue to set teaching objectives, select materials for use and make assessment decisions.

4. Comparison with teaching workflows without AI

Teaching task	Typical workflow without AI	Proposed contribution of Lumori
Prepare materials	Select existing resources; write and adapt course explanations.	Generate editable drafts from course sources and learning objectives.
Design practice	Reuse or write questions and solutions; adapt them to learner needs.	Generate questions by format and intended difficulty; revise individual items.
Check and revise	Find references, verify answers and maintain document versions.	View generation sources, edit items and retain review and version records.
Support self-study	Read notes, use available exercises and seek help from educators or peers.	Request explanations and practice; optionally ask follow-up questions by voice.
Present content	Prepare text, recordings and visual aids separately.	Generate audio and visual aids from reviewed text, then check each output.

Expected benefits: less repetitive drafting and easier checking for educators; more accessible practice and explanations for students. These are hypotheses: total effort must include reviewing and correcting AI output.

5. How the value will be evaluated

Educator workflow: compare manual preparation, general-purpose chat AI and Lumori on matched tasks. Measure total time including review, correction effort and final quality.

Content quality: seek independent subject-informed ratings of correctness, source support, answerability and intended difficulty. Record failures; actual student difficulty requires learner response data.

Media and interaction: check cross-modal consistency. If voice discussion is built, compare it with text for clarification time, comprehension and preference. Improved grades require a separate learner study.

6. Delivery boundaries and current position

The core scope excludes a full learning management system, automated grading, secure examination delivery, long-term adaptive learner modelling, video generation and training a foundation model from scratch.

The current project provides the text-authoring and review foundation, with speech/image adapters. End-to-end multimodal quality and the proposed user benefits still require evaluation; voice discussion is a proposed extension. Accounts currently own their materials; cross-account classroom distribution is not implemented.

Brief alignment. The original brief specifies content and assessment generation, RAG, educator/student interfaces and at least two alternative modalities. CA1 focuses on the text/RAG prototype; CA2 integrates other modalities. Source: *CEG 311 FYP Project Brief, AY 2026/2027*, p. 27, "AI-Powered Multimodal Content & Assessment Generation for Educational Applications."